EXPERIMENT 1 – COMPARATIVE STUDY OF LOW-CODE DEVELOPMENT VERSUS TRADITIONAL APPLICATION DEVELOPMENT APPROACHES.

Name: Bakkialakshmy P

Roll no: 240701067

Aim:

Study and compare the traditional software development approach with the low-code development approach To and understand their advantages, disadvantages, development process, and application in modern software development.

Introduction to Traditional Approach:

What is Traditional Application Development?

Traditional application development is the process of building software by writing programming code manually. Developers use programming languages like Java, Python, C#, PHP, or JavaScript to create applications.

In this approach, every feature such as forms, database connections, validations, reports, and user interfaces must be coded by developers.

It requires strong programming knowledge, testing, debugging, and longer development time.

Features:

- Manual coding
- Requires programming knowledge
- More development time
- Higher maintenance effort
- Suitable for complex applications

Introduction to Low-Code Development:

What is Low-Code Development?

Low-code development is a modern software development approach that allows applications to be built using drag-and-drop tools with very little coding.

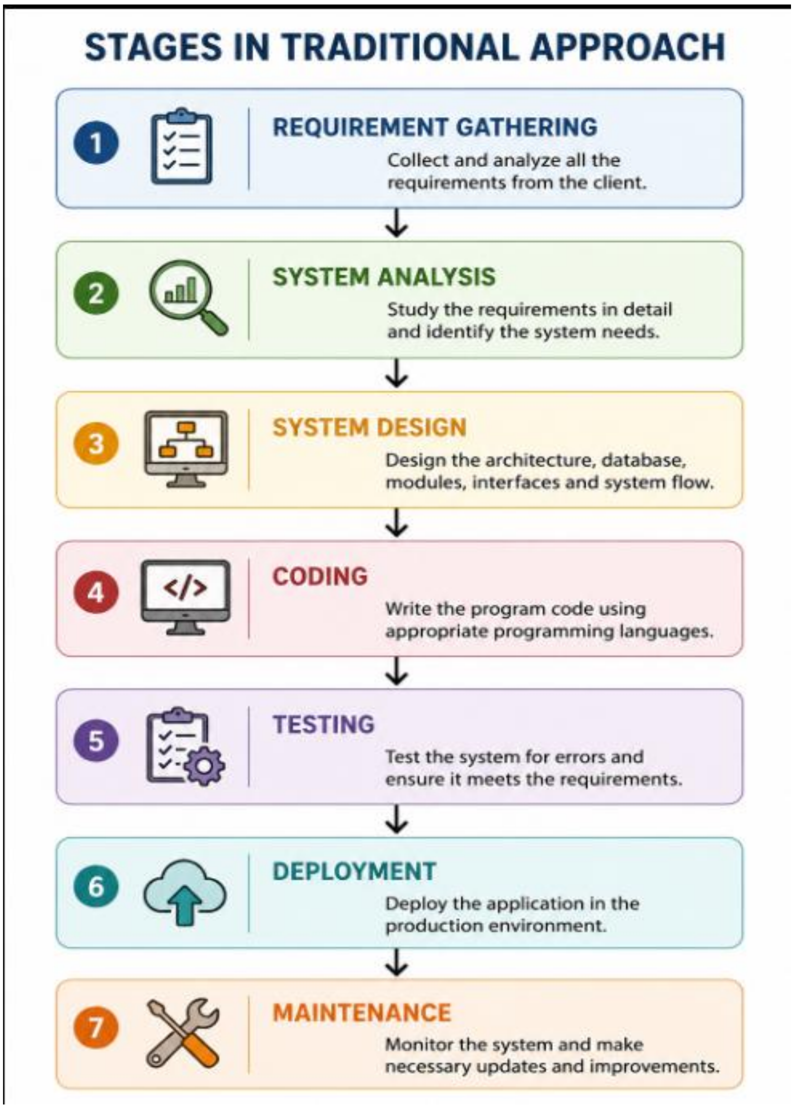
Platforms like Zoho Creator, Microsoft Power Apps, OutSystems, and Mendix help developers create applications quickly.

Only a small amount of code is required for advanced logic.

Features:

- Drag-and-drop interface
- Minimal coding
- Faster development
- Easy maintenance
- Suitable for business applications

Diagram for Stages in Traditional Approach:



Explanation:

1. Requirement Gathering – Collect customer requirements.
2. Analysis – Study the requirements.
3. Design – Prepare database and software design.
4. Coding – Write the program.
5. Testing – Find and fix errors.
6. Deployment – Release the software.
7. Maintenance – Update and support the application.

Diagram for Low-Code Development:

Explanation:

- Identify business requirements.
- Create a blank application.
- Design forms using drag-and-drop.
- Configure workflows and automation.
- Generate reports and dashboards.
- Test the application.
- Publish for users.

Comparative Analysis:

Comparative Analysis of Traditional and Low-Code Development

S.No.	Comparison Parameter	Traditional Application Development	Low-Code Development
1.	**Development Approach**	Manual coding using programming languages	Visual development with drag-and-drop components
2.	**Coding Requirement**	Requires extensive coding	Requires minimal coding
3.	**Development Speed**	Slow	Fast
4.	**Programming Skills**	Advanced programming knowledge required	Basic programming knowledge is sufficient
5.	**Development Cost**	High	Low
6.	**User Interface Design**	Developed manually through coding	Designed using drag-and-drop tools
7.	**Database Management**	Created and managed manually	Automatically generated and managed
8.	**Testing**	Mostly manual testing	Built-in testing and validation support
9.	**Maintenance**	More time and effort required	Easier and quicker to maintain
10.	**Deployment**	Manual deployment process	One-click or simplified deployment
11.	**Flexibility**	Highly customizable	Limited to platform features with scripting support
12.	**Examples**	Java, Python, C#, PHP, .NET	Zoho Creator, Microsoft Power Apps, Mendix, OutSystems

Result:

The experiment was successfully completed. The traditional application development approach and the low-code development approach were studied and compared. It was observed that low-code development reduces coding effort, speeds up application development, and simplifies maintenance, while traditional development provides greater flexibility and is suitable for highly complex software systems.